Chapter 8, Lecture 9

Marten Wegkamp

Normal theory

Throughout, we assume that we sample from a Normal distribution: $X_1, \ldots, X_n$ are iid $N(\mu, \sigma^2)$.

Claim: $\overline{X}_n$ and S^2 are independent.

Proof: Set

$$Z_i = (x_i - \mu)/\sigma \sim N(0,1).$$

Then

$$\overline{Z}_n = \frac{\overline{X}_n - \mu}{\sigma} \qquad \text{and} \qquad \sum_{i=1}^n (Z_i - \overline{Z}_n)^2 = \sum_{i=1}^n (X_i - \overline{X}_n)^2/\sigma^2.$$

Consider n orthonormal vectors $A_1, \ldots, A_n$ in $\mathbb{R}^n$, with

$$A_1 := (n^{-1/2}, \cdots, n^{-1/2})^\top$$

and set

$$Y_i := \sum_{j=1}^n A_{ij} Z_j.$$

We will show that $Y_1, \ldots, Y_n$ are iid $N(0,1)$. Indeed, Y_i are Normal with mean

$$\mathbb{E}[Y_i] = \sum_{j=1}^n A_{ij} \mathbb{E}[Z_j] = 0$$

and variance

$$\mathrm{Var}(Y_i) = \sum_{j=1}^n A_{ij}^2 \mathrm{Var}(Z_j) = \sum_{j=1}^n A_{ij}^2 = 1.$$

Since the Y_i are normal, independence follows from covariance equal to zero, and indeed,

$$\begin{aligned} \mathrm{Cov}(Y_i, Y_j) &= \sum_{k=1}^n \sum_{\ell=1}^n A_{ik} A_{j\ell} \mathrm{Cov}(Z_k, Z_\ell) \\ &= \sum_{k=1}^n A_{ik} A_{jk} \\ &= 0 \qquad \text{for } i \neq j \text{ as } A_i \perp A_j. \end{aligned}$$

This means in particular that

$$Y_1 = \sum_{j=1}^{n} A_{1j} Z_j = \sqrt{n}\overline{Z}_n = \sqrt{n}\frac{\overline{X}_n - \mu}{\sigma}$$

is independent of $\sum_{i=2}^{n} Y_i^2$. Use $\mathbf{Z} = (Z_1, \ldots, Z_n)^\top$, $\mathbf{Y} = \mathbf{AZ} = (Y_1, \ldots, Y_n)^\top$ with $\mathbf{A}$ the $n \times n$ matrix with row vectors $A_1, \ldots, A_n$. Hence, $\mathbf{Y}$ is a rotation of $\mathbf{Z}$, and we just showed that the normal distribution is rotation invariant.

$$\begin{aligned}
\sum_{i=2}^{n} Y_i^2 &= \sum_{i=1}^{n} Y_i^2 - n\overline{Z}_n^2 \\
&= \mathbf{Y}^\top \mathbf{Y} - n\overline{Z}_n^2 \\
&= \mathbf{Z}^\top \mathbf{A}^\top \mathbf{A}\mathbf{Z} - n\overline{Z}_n^2 \\
&= \mathbf{Z}^\top \mathbf{Z} - n\overline{Z}_n^2 \qquad \text{as } \mathbf{A}^\top \mathbf{A} = I_n \\
&= \sum_{i=1}^{n} (Z_i - \overline{Z}_n)^2 \\
&= (n-1)S^2/\sigma^2
\end{aligned}$$

This implies that $\overline{X}_n$ and S^2 are independent. ■

Claim: $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Proof. Since

$$\frac{n-1}{\sigma^2} S^2 = \frac{1}{\sigma^2} \sum_{i=1}^{n} (X_i - \overline{X}_n)^2 = \sum_{i=2}^{n} Y_i^2$$

and $Y_2, \ldots, Y_n$ are iid $N(0,1)$, the result follows immediately. ■

Claim: $\sqrt{n}(\overline{X}_n - \mu)/S \sim t_{n-1}$.

Proof: Note that

$$\sqrt{n}\frac{\overline{X}_n - \mu}{S} = \frac{\sqrt{n}\overline{Z}_n}{\sqrt{S^2/\sigma^2}} = \frac{Y_1}{\sqrt{\sum_{i=2}^{n} Y_i^2/(n-1)}}$$

and the result follows. ■

Exact confidence interval for μ

Since $\sqrt{n}(\overline{X}_n - \mu)/S \sim t_{n-1}$, we easily find the exact confidence interval

$$[\overline{X}_n \pm \frac{S}{\sqrt{n}} t_{n-1}(\alpha/2)]$$

Here $t_{n-1}(\alpha/2)$ is the $\alpha/2$ upper point of t_{n-1}. This is typically a wider confidence interval than the asymptotic interval using $z(\alpha/2)$ in lieu of $t_{n-1}(\alpha/2)$.

Exact confidence interval for σ^2

Let $k_n(\alpha)$ be the α-upper point of a chi-square χ^2_n. Since

$$(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$$

we find

$$\begin{aligned} 1-\alpha &= \mathbb{P}\left\{k_{n-1}(1-\alpha/2) \leq \frac{(n-1)S^2}{\sigma^2} \leq k_{n-1}(\alpha/2)\right\} \\ &= \mathbb{P}\left\{\frac{(n-1)S^2}{k_{n-1}(\alpha/2)} \leq \sigma^2 \leq \frac{(n-1)S^2}{k_{n-1}(1-\alpha/2)}\right\} \end{aligned}$$

so that

$$\left[\frac{(n-1)S^2}{k_{n-1}(\alpha/2)}, \frac{(n-1)S^2}{k_{n-1}(1-\alpha/2)}\right]$$

is an exact $1-\alpha$ confidence interval for σ^2.